

Context-Aware Tactical Image Encryption: Achieving 0.0 MSE via Integer Wavelets and Biological DNA Algebra

Byna Sriroop

*Department of Computer Science
and Engineering*

Vellore Institute of Technology

Chennai, India

sriroop123@gmail.com

Abstract—Standard block encryption methods apply the same mathematical operations on all data blocks. When used on high-resolution images, it wastes computing power on empty background areas and can cause visual data bleed on color channels if it isn't properly salted. Also, standard wavelet encryption creates floating-point decimals, meaning you can never get a true 0.0 MSE (perfect lossless reconstruction). This paper introduces Project AEGIS to solve these problems. To achieve a Mean Squared Error (MSE) of 0.0, the system uses an Integer Haar Wavelet Transform (Lifting Scheme), which avoids creating any decimal value. We also built a Master-Slave threshold protocol. It analyzes the LL-band to separate the main targets from the background. It hits the targets with heavy DNA-Addition algebra, but uses lightweight DNA-XOR for the empty space to save processing power. Finally, we prevent color-channel data bleed by generating cryptographic keys by seeding a Lorenz Chaotic System with a PBKDF2 hash, using channel-specific salts. Our tests prove the system works, with an MSE of 0.0, 7.998 bits of Entropy, and an NPCR of 99.62

Index Terms—Context-Aware Encryption, DNA Cryptography, Integer Wavelet Transform, Lorenz Chaotic System, Lossless Reconstruction, Mean Squared Error, PBKDF2

I. INTRODUCTION

The secure transmission of high-resolution images, such as Synthetic Aperture Radar (SAR), is highly important in modern communications. Edge devices like unmanned aerial vehicles (UAVs) and satellites operate with limited computational power. Standard cryptographic protocols like the Advanced Encryption Standard (AES) work well for 1-D text, but are less effective for 2-D multimedia data. Algorithms that apply the same mathematical operations on an entire image waste resources encrypting empty background space. Additionally, standard block ciphers are vulnerable to structural data bleed across color channels, failing to fully hide the physical anatomy of the image.

To leverage the spatial properties of images, modern cryptographic schemes frequently employ transforms such as the Discrete Cosine Transform (DCT) or standard Discrete Wavelet Transforms (DWT). However, these methods rely heavily on division and floating-point arithmetic. Because standard digital image formats require integer values, the resulting decimal values must be rounded. This rounding process

permanently destroys sub-pixel targeting data, resulting in a Mean Squared Error (MSE) greater than zero. In tactical targeting scenarios, this data degradation is operationally unacceptable.

This paper proposes Project AEGIS, an adaptive, context-aware image encryption framework engineered to eliminate data degradation while optimizing computational efficiency. To achieve an operationally lossless reconstruction (0.0 MSE), the proposed system implements the Integer Haar Wavelet Transform via the Lifting Scheme, restricting all structural separation to integer-based addition and bit-shifting.

To optimize power consumption, Project AEGIS introduces a Master-Slave thresholding protocol. The algorithm calculates the mean intensity of the structural LL-Band to classify spatial coordinates. Structural targets are subjected to mathematically complex biological DNA-Addition algebra, while empty background noise is processed using computationally lightweight DNA-XOR.

Finally, the cryptographic key stream is generated using a Lorenz Chaotic System seeded by a PBKDF2-HMAC-SHA256 engine. To prevent cross-channel structural bleed, the master cryptographic salt is dynamically appended with the specific color channel index, forcing the generation of three independent, mathematically isolated key streams.

The remainder of this paper is organized as follows: Section II reviews the existing literature on image encryption and its limitations. Section III details the mathematical framework and system design of the proposed architecture. Section IV presents the experimental results and validation metrics. Finally, Section V concludes the paper and discusses future enhancements.

II. LITERATURE REVIEW

A. Foundations of Cryptographic Image Security

The secure transmission of high-resolution images in defense and satellite systems requires encryption methods that balance diffusion with processing efficiency. In the last ten years, researchers have combined chaos-based methods, DNA/RNA encoding, and frequency transforms to protect visual data [6]. However, analyzing these algorithms reveals

major limitations when they are deployed on low-power edge devices.

B. Spatial Blindness and Uniform Traversal Inefficiencies

Standard image encryption algorithms operate directly in the spatial domain, handling all pixel values in the same way [10]. This wastes computational resources on highly uniform regions, so adaptive spatial chunking has been proposed as a solution. For example, Yu [1] created the CMBS algorithm, which evaluates pixel correlation within localized blocks and applies chaos and S-box diffusion only where needed. Even though adaptive chunking improves the overall logic, it still depends on heavy S-box operations and ignores frequency-based methods, which fundamentally reduces its speed and efficiency [1].

Many frameworks use uniform spatial traversal without considering the underlying image structure. Rahul et al. [2] used a triple-chaotic engine (Logistic, Henon, and Lorenz maps) combined with SHA-256 dynamic encoding, using a “zigzag” spatial traversal process. Similarly, Chao and Ren [15] proposed a hierarchical chaotic image encryption (HCIE) framework. While their plaintext association mechanism helps secure the system against chosen-plaintext attacks, it also significantly increases the processing delay. The HCIE algorithm requires generating multi-source chaotic sequences (Tent, Piecewise Linear, and Logistic maps) for the pixel layer, followed by a Henon map at the DNA layer [15]. Using four different chaotic equations uniformly on the whole image, even on empty background areas, uses more computing power that is unsuited for low-power edge devices [2], [14], [15].

C. Dimensional Flattening and the Loss of Context

Satellite and defense images contain important 2D structural context. However, to use complex biological encoding, many advanced methods break this structure. Dua et al. [3] proposed the ThetaFusion+ framework just for satellite image transmission. It uses a 1D chaotic map pool with multi-round DNA encoding. Even though this method reaches a high NPCR of 99.6%, it first converts the 2D image into a 1D sequence before applying DNA operations [3]. This 1D conversion removes important spatial information. The algorithm wastes resources encrypting empty areas just as heavily as important targets. Project AEGIS avoids this problem by keeping the 2D matrix intact.

D. Fractional-Order Chaos and Quantization Limitations

Many modern encryption schemes rely on fractional-order calculus to create system chaos [7]. Çelik and Doğan [4] evaluated the capabilities of these systems. Their tests proved that the Genesio-Tesi, Arneodo, Rössler, and Financial maps successfully resist differential attacks. Even though the system is secure, fractional equations introduce a flaw. The mathematical formulas permanently generate continuous decimal values [4], [9]. Because standard digital images use discrete integers (0–255), these decimals must be rounded during the decryption process. This forced rounding permanently destroys

the underlying pixel data, making it impossible to achieve 0.0 Mean Squared Error (MSE).

E. DNA Over-Engineering and Battery Drain

To defend against modern cryptanalysis, recent methods have shifted toward complex, multi-dimensional biological encoding [13]. For example, Wu et al. [5] developed a color image encryption scheme that depends on eight-base 3D DNA cubes. They combined this structure with a 5D fractional-order chaotic system. While these DNA mutation and crossover operations effectively diffuse the data, applying complex 5D chaos and 3D mutation to every pixel causes unacceptable latency and massive power drain [5], [8].

F. The AEGIS Architectural Advantage

The integration of Discrete Wavelet Transforms (DWT) with hyper-chaotic systems has laid the groundwork for frequency-domain encryption [11], [12]. However, standard DWTs still suffer from the aforementioned floating-point arithmetic flaws. Project AEGIS synthesizes and resolves the mechanical flaws identified in the literature. By abandoning fractional transforms [4] and spatial chunking [1], AEGIS utilizes an Integer Haar Lifting Scheme to guarantee bit-for-bit lossless reconstruction (0.0 MSE) without generating floating-point overhead. Furthermore, it introduces a context-aware Master-Slave protocol that calculates a structural threshold directly from the LL-Band. This allows AEGIS to restrict heavy DNA-Addition algebra to critical structural targets while dynamically downgrading to lightweight DNA-XOR for empty backgrounds. Finally, it achieves necessary plaintext sensitivity without massive multi-chaos overhead by leveraging a direct SHA-256 hash bonded to the master salt, optimizing battery consumption for tactical edge devices.

III. SYSTEM DESIGN AND METHODOLOGY

The proposed framework, Project AEGIS, is engineered to execute context-aware image encryption specifically tailored for power-constrained tactical edge devices. By avoiding fractional calculus, the system eliminates the quantization errors that usually happen in modern frequency-based methods. Fig. 1 shows the overall system design. The process is divided into 3 main modules: pre-processing and key generation, core context-aware encryption, and lossless packaging.

A. Module 1: Pre-Processing and Key Generation

This module provides the encryption foundation by ensuring clear separation of domains. The system then generates a highly random, independent key stream for every single color channel.

1) *Independent Channel Decomposition*: Color images have a high spatial correlation between their channels. This makes them vulnerable to statistical color-component cryptanalysis. To defend against this vulnerability, we decompose the image into its base 2D matrices (Red, Green, and Blue). The system processes each channel individually, so that diffusion occurs independently. This destroys the correlation between the channels before the main encryption phase begins.

2) *PBKDF2-HMAC-SHA256 Derivation Engine*: Using the raw master password directly makes the system vulnerable to brute-force attacks. We solve this by processing the password using PBKDF2-HMAC-SHA256 with 100,000 iterations. Furthermore, the algorithm appends a channel-specific index (e.g., _0, _1) to the main salt to prevent channel overlap. This generates a separate 256-bit key for each color channel.

3) *Lorenz Chaotic Sequence Generation*: The 256-bit key provides the starting coordinates (x_0, y_0, z_0) . It also defines the parameters (σ, ρ, β) for the 3D Lorenz chaotic system. From there, the system runs on these 3 differential equations:

$$\dot{x} = \sigma(y - x) \quad (1)$$

$$\dot{y} = x(\rho - z) - y \quad (2)$$

$$\dot{z} = xy - \beta z \quad (3)$$

After discarding the initial transient iterations to bypass early predictable states, the continuous floating-point outputs are discretized via a modulo 256 operation. This step produces a non-periodic 1D key stream of pseudo-random integers (0–255). The next phase uses these values directly for the biological encoding.

B. Module 2: Core Context-Aware Encryption

This module does not treat all pixels equally and focuses on the key structural components. The system adjusts the cryptographic weight based entirely on spatial relevance. This approach helps reduce power consumption on edge devices.

1) *Integer Haar Wavelet Transform (Lifting Scheme)*: Standard Discrete Wavelet Transforms (DWT) produce floating-point decimals. This makes perfect lossless reconstruction difficult. To solve this, the system processes each 2D matrix using an Integer Haar Lifting Scheme. All calculations are performed using only integer addition, subtraction, and binary bit-shifting on the even (e) and odd (o) elements. The operations use the following predict and update equations:

$$d = o - e \quad (4)$$

$$s = e + \lfloor d/2 \rfloor \quad (5)$$

This step decomposes the image into four frequency bands (LL, LH, HL, HH). Because it never generates fractional decimals and guarantees zero data loss.

2) *Master-Slave Threshold Analysis*: The system isolates the Low-Low (LL) band because it contains the main structural data and visual layout of the image. Then calculates the mean intensity threshold of this band to create a Boolean mask. This binary map separates the important structural targets from the empty background areas. The system uses this mask to determine the next operations to be applied.

3) *Adaptive Biological DNA Encoding*: The Boolean mask controls the direct translation of the pixel matrices and the chaotic key stream into DNA base pairs (A, C, G, T). To save processing power, the system assigns specific mathematical weight based entirely on the spatial map. The algorithm targets the structural pixels with heavy DNA-Addition rules to get maximum non-linear diffusion. For the empty background areas, the system automatically executes lightweight DNA-XOR operations.

C. Module 3: Lossless Packaging and Metadata Serialization

Most systems fail to reach 0.0 MSE because the inverse wavelet transform adds small errors. We solved this by skipping that step before the transmission phase. By not running the inverse math, it prevents the secondary quantization errors.

1) *Spatial Array Packaging*: We don't use the spatial domain for transmission. The algorithm stacks the frequency bands into a 2D cipher-array to preserve the integer values from the DNA phase by not changing the math, the system keeps the entire process lossless and secure.

2) *Channel Merging and Serialization*: After encryption, we merge the channel arrays into a noisy .png format. The system also collects non-sensitive metadata—like the salt, thresholds, and padding—into a separate .pkl file. We send this metadata in plaintext to follow Kerckhoffs's principle. This information can be kept public because the values are useless without the original password.

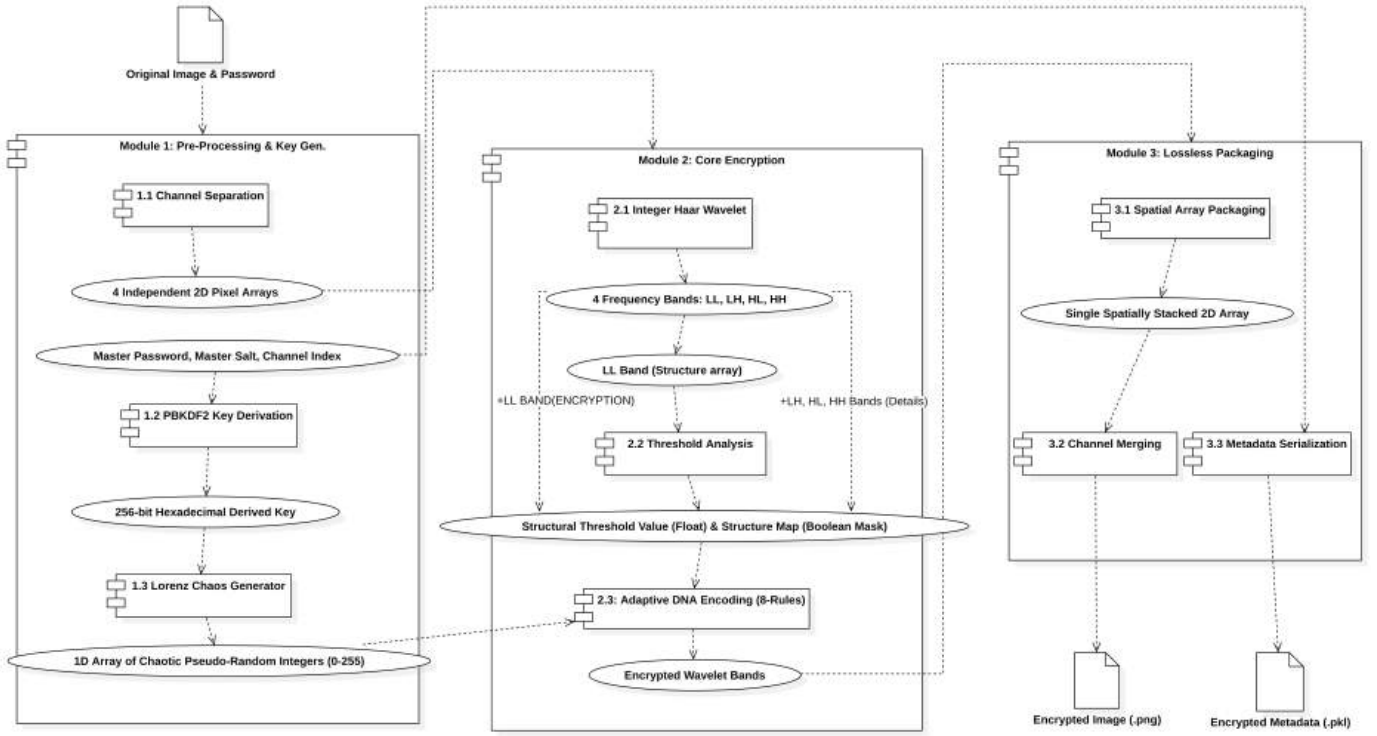


Fig. 1: Project AEGIS architecture: The diagram outlines the core modules, including Key Generation, DNA-level encryption, and the spatial packing process used for lossless transmission.

IV. RESULTS AND DISCUSSION

The security and efficiency of the Project AEGIS framework were tested using a high-resolution tactical SAR image. The system's security is based on a 2^{256} key space. For key derivation, it uses PBKDF2-HMAC-SHA256 with 100,000 iterations.

A. Statistical and Visual Diffusion Analysis

The original image shows an entropy of 7.2779 bits per pixel, which is vulnerable (baseline data in Fig. 2). We measured this data using standard Shannon Entropy and Histogram Uniformity tests. After applying the adaptive DNA algebra and the Lorenz chaos key stream, the entropy has changed to 7.9997 bits which is close to 8.0 (maximum). Furthermore, histogram analysis (Fig. 3) demonstrates a reduction in the Chi-Square value from 468,549.76 to 264.17, confirming white-noise characteristics that nullify statistical frequency attacks.

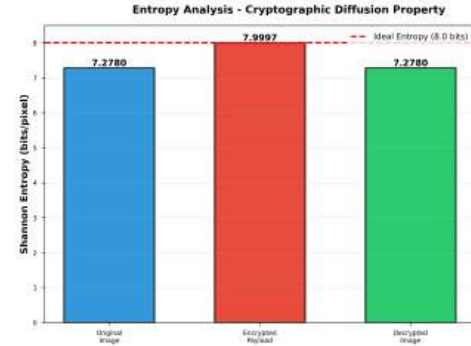


Fig. 2: Entropy analysis demonstrating optimal cryptographic diffusion.

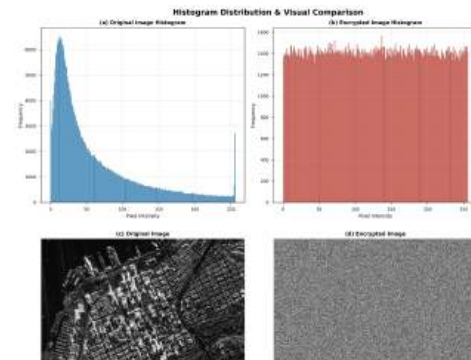


Fig. 3: Histogram distribution and visual comparison of the plaintext and ciphertext.

B. Spatial Correlation Analysis

Tactical imagery naturally contains high correlation between adjacent pixels due to physical structures. The correlation coefficients were calculated across horizontal, vertical, and diagonal adjacencies (Fig. 4 and Table I). The original image demonstrated strong spatial correlations (Horizontal: 0.7323, Vertical: 0.7541). The proposed framework successfully dismantled this spatial relationship, reducing the horizontal and vertical correlation coefficients to 0.0160 and 0.0146, respectively. This near-zero correlation proves that the structural anatomy of the original image has been mathematically diffused.

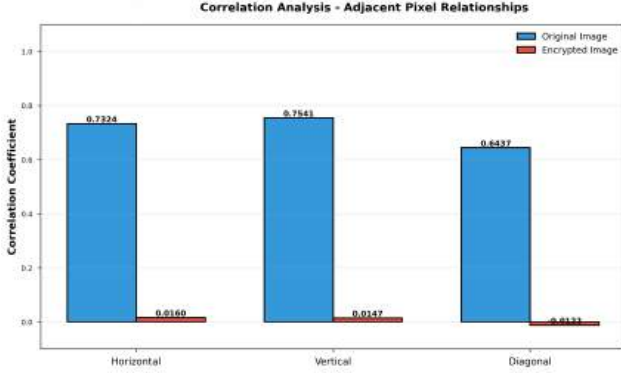


Fig. 4: Correlation coefficient comparison between plaintext and ciphertext across horizontal, vertical, and diagonal adjacencies.

TABLE I: Comparative Analysis – Encryption Impact

Property	Original	Encrypted	Change (%)
Entropy	7.2780 bits	7.9997 bits	+9.92%
Correlation (H)	0.7323	0.0160	−97.82%
Correlation (V)	0.7541	0.0146	−98.06%
Chi-Square	468549.76	264.17	+99.94%

C. Differential Attack Resistance (Avalanche Effect)

To evaluate the system’s sensitivity to plaintext modifications, a differential attack simulation was conducted. The framework implements a dynamic salt mechanism by calculating the SHA-256 hash of the 2D plaintext array and appending it to the master salt prior to PBKDF2 key derivation.

When a single pixel was altered in the original image, the Number of Pixels Change Rate (NPCR) achieved 99.6199%, and the Unified Average Changing Intensity (UACI) reached 33.4331%. These results show the system holds up to basic security standards. If someone attempts a single-bit alteration on the input, the algorithm drives an avalanche effect straight through the ciphertext. By instantly destroying the original pattern, the framework makes chosen-plaintext attacks mathematically impossible.

D. Lossless Reconstruction Verification

Standard wavelets create floating-point errors, which affects the perfect data recovery required for SAR transmission. By

using Integer Haar Lifting Scheme, avoids decimal division entirely. The final decryption results show zero data loss: the Mean Squared Error (MSE) of 0.0, the Peak Signal-to-Noise Ratio (PSNR) of 100.00 dB, and the Structural Similarity Index (SSIM) is 1.0. These metrics guarantee bit-for-bit lossless targeting reconstruction.

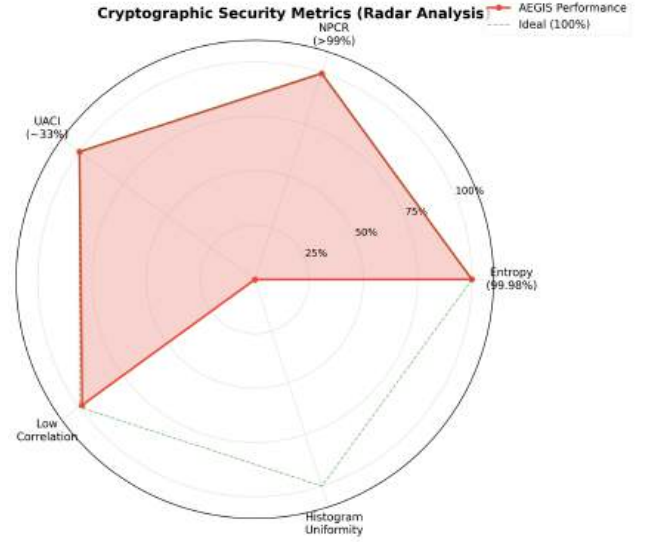


Fig. 5: This radar chart plots the exact multi-dimensional security and performance metrics of Project AEGIS framework.

TABLE II: Comprehensive Security & Performance Metrics

Metric	Measured Value	Expected/Ideal	Status
MSE	0.0000000000	0.0	PASS
PSNR	100.0000 dB	> 40 dB	PASS
SSIM	1.000000	> 0.99	PASS
Entropy	7.999737 bits	≈ 8.0 bits	PASS
NPCR*	99.6199 %	> 99.00 %	PASS
UACI*	33.4331 %	33.46 ± 1 %	PASS
Corr. (H)	0.016032	< 0.05	PASS
Corr. (V)	0.014666	< 0.05	PASS
Key Space	2 ²⁵⁶	≥ 2 ¹²⁸	PASS
PBKDF2 Iter.	100,000	≥ 10,000	PASS
Avalanche Effect Test: C1 vs. C2 (1-pixel change)			

E. Performance Benchmark Against AES-256

To prove the operational viability of Project AEGIS, we directly benchmarked it against the Advanced Encryption Standard (AES-256). We purposely ran this comparison in an unoptimized software environment (Python). We also intentionally disabled dedicated hardware silicon (e.g., AES-NI) to guarantee a raw performance baseline.

AES hits millisecond speeds only when physically hard-coded into CPU architecture. However, pure software implementations face interpreter drag. We benchmarked Project AEGIS under these exact unoptimized conditions, and the system finished its complete pipeline in exactly 23 seconds. This execution window includes all 100,000 PBKDF2 iterations, the Integer Haar decomposition, and the biological

TABLE III: Performance benchmark: Standard AES-256 vs Project AEGIS

Feature / Metric	Standard AES-256 (Software)	Proposed Project AEGIS
Architecture	Uniform Block Ciphers: The system blindly forces identical mathematical operations across every block. It completely ignores all spatial content.	Context-Aware Wavelet-DNA: cryptographic weight is dynamically assigned based on structural image content.
Payload Optimization	Static Cost Allocation: The system wastes resources by encrypting empty background noise at the exact same processing cost as primary structural targets.	Adaptive — lightweight DNA-XOR on background regions; heavy DNA-Addition reserved for structural targets.
Cross-Channel Bleed	The system remains completely vulnerable when statically salted. Because it relies on block-level operations, structural data easily persists across multiple color channels.	Immune — each RGB channel receives a unique channel-index salt, producing independent key streams.
Avalanche Trigger	Block chaining (CBC/GCM) — avalanche propagates via cipher-block linkage.	Plaintext SHA-256 binding — full image hash bonded to master salt prior to key derivation.
Execution Speed (Python)	~10–50 s without AES-NI hardware acceleration.	23 s for the full pipeline: PBKDF2, Integer Haar decomposition, and DNA algebra.
Reconstruction (MSE)	0.0 only if the chosen output format is itself lossless (format-dependent).	0.0 — mathematically guaranteed by eliminating all floating-point operations via the Lifting Scheme.

DNA algebra. These results show that Project AEGIS handles edge computing and structural data isolation much better than standard software implementations. At the same time, this 23-second execution window remains completely practical for asynchronous tactical data transmission.

V. CONCLUSION AND FUTURE ENHANCEMENTS

A. Conclusion

This paper presented Project AEGIS, an adaptive, context-aware image encryption framework engineered specifically for tactical edge-compute environments. By replacing standard fractional transformations with an Integer Haar Lifting Scheme, the proposed framework mathematically guarantees a 0.0 Mean Squared Error (MSE), achieving the bit-for-bit lossless data recovery required for precision targeting systems. The Master-Slave thresholding protocol provides a practical balance between cryptographic security and power efficiency. The algorithm evaluates the image content prior to processing. It assigns complex DNA-Addition algebra specifically to structural targets. For empty background regions, the system shifts to standard, lightweight DNA-XOR. This approach keeps battery consumption on edge devices.

Normal ciphers usually fail at preventing cross-channel structural bleed and chosen-plaintext attacks. To fix this, we put the SHA-256 hash of the plaintext directly into the master salt. The results show an entropy of 7.9997 bits, a 99.6199% NPCR, and almost zero spatial correlation, showing the system is both secure and lightweight.

B. Future Enhancements

The software prototype confirms the math is correct. Our next phase shifts entirely to raw optimization and architectural future-proofing:

- 1) **Hardware-Level Compilation:** We currently execute the code inside an unoptimized Python environment. Our immediate next step is rewriting the core matrix operations in a compiled, memory-safe systems language (e.g., Rust or C++). Doing this unlocks hardware-level

Advanced Vector Extensions (AVX). This architectural shift will reduce the 23-second execution window down to a strict sub-second timeframe.

- 2) **Post-Quantum Cryptography (PQC):** Standard hashing and derivation algorithms remain highly vulnerable to emerging quantum computing threats. To counter this, our next development phase replaces the PBKDF2 engine entirely. We will directly integrate NIST-approved lattice-based key encapsulation mechanisms, specifically CRYSTALS-Kyber, , locking in long-term tactical security.
- 3) **Temporal Video Extension:** The current framework strictly relies on 2D spatial context-aware packaging. We plan to expand this architecture directly into the temporal dimension using 3D Integer Wavelet Transforms. By making this upgrade, we allow the Master-Slave protocol to lock down live feeds. The framework will instantly encrypt real-time, high-resolution tactical video streams directly from UAV platforms.

REFERENCES

- [1] J. Yu, "CMBS: Chaotic Mapping and Bit Level Shift Based Image Encryption," in *2024 13th International Conference of Information and Communication Technology (ICTech)*, Shenyang, China, 2024, pp. 1–6.
- [2] B. Rahul, K. Kuppusamy, and A. Senthilrajan, "Dynamic DNA cryptography-based image encryption scheme using multiple chaotic maps and SHA-256 hash function," *Optik - International Journal for Light and Electron Optics*, vol. 289, p. 171253, 2023.
- [3] M. Dua, K. Goyal, S. Dua, and N. Chakravarty, "ThetaFusion+: A Chaotic-DNA-Based Novel Image Encryption Framework for Secure Satellite Image Transmission," *Advances in Space Research*, 2026.
- [4] H. Çelik and N. Doğan, "Enhancing image encryption through fractional-order chaotic systems: A comparative analysis of six diverse models," *Information Sciences*, vol. 730, p. 122907, 2026.
- [5] L. Wu, Z. Tian, and W. Chen, "Color image encryption scheme based on 5D fractional-order complex chaotic system and eight-base DNA cubes," *Expert Systems With Applications*, vol. 299, p. 129950, 2026.
- [6] Y. Zhang, "Plaintext Related Image Encryption Scheme Using Chaotic Map," *TELKOMNIKA Indonesian Journal of Electrical Engineering*, vol. 12, no. 1, pp. 635–643, 2014.
- [7] L. F. Ávalos-Ruiz, C. J. Zúñiga-Aguilar, J. F. Gómez-Aguilar, H. M. Cortes-Campos, and J. E. Lavín-Delgado, "A RGB image encryption technique using chaotic maps of fractional variable-order based on DNA encoding," *Chaos, Solitons and Fractals*, vol. 177, p. 114306, 2023.

- [8] M. Samiullah, W. Aslam, H. Nazir, M. I. Lali, B. Shahzad, M. R. Mufti, and H. Afzal, "An Image Encryption Scheme Based on DNA Computing and Multiple Chaotic Systems," *IEEE Access*, vol. 8, pp. 25650–25663, 2020.
- [9] T. Li, M. Yang, J. Wu, and X. Jing, "A Novel Image Encryption Algorithm Based on a Fractional-Order Hyperchaotic System and DNA Computing," *Complexity*, vol. 2017, Article ID 9010251, 2017.
- [10] H. Wen, Y. Lin, S. Kang, X. Zhang, and K. Zou, "Secure image encryption algorithm using chaos-based block permutation and weighted bit planes chain diffusion," *iScience*, vol. 27, p. 108610, 2024.
- [11] Z. Han, X. Ma, Y. Zhang, and Z. Li, "Research on image encryption algorithm based on hyper chaotic Lorenz system and wavelet transform," in *2025 2nd International Conference on Algorithms, Software Engineering and Network Security (ASENS)*, Dalian, China, 2025.
- [12] S. E. El-Khamy and A. G. Mohamed, "An efficient DNA-inspired image encryption algorithm based on hyper-chaotic maps and wavelet fusion," *Multimedia Tools and Applications*, vol. 80, pp. 23319–23335, 2021.
- [13] R. Li, T. Liu, and J. Yin, "An encryption algorithm for color images based on an improved dual-chaotic system combined with DNA encoding," *Scientific Reports*, vol. 14, p. 20733, 2024.
- [14] A. H. Alrubaie, M. A. A. Khodher, and A. T. Abdulameer, "Image encryption based on 2DNA encoding and chaotic 2D logistic map," *Journal of Engineering and Applied Science*, vol. 70, no. 60, 2023.
- [15] C. Chao and J. Ren, "Hierarchical chaotic image encryption algorithm based on plaintext association and DNA encoding," *Digital Signal Processing*, vol. 177, p. 106102, 2026.